

PRACTICAL 3: To group datasets based on categorical attributes and compute aggregate statistics such as mean, sum, count, and percentage.

Input: Hospital_Patient_Analytics_4000.csv

Code:

```
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt

# Load Dataset
DATA_FILE = Path(__file__).resolve().parent / "Hospital_Patient_Analytics_4000.csv"
df = pd.read_csv(DATA_FILE)

# Q1. Which hospital generated the highest revenue and
#    what factors might explain this?

hospital_analysis = df.groupby("Hospital").agg(
    Total_Revenue=("Treatment_Cost", "sum"),
    Total_Patients=("Patient_ID", "count"),
    Average_Treatment_Cost=("Treatment_Cost", "mean"),
    Average_Medicine_Cost=("Medicine_Cost", "mean"),
    Average_Lab_Cost=("Lab_Test_Cost", "mean"),
    Average_Surgery_Cost=("Surgery_Cost", "mean"),
    Average_Hospital_Stay=("Hospital_Stay_Days", "mean"),
    Average_Satisfaction=("Patient_Satisfaction", "mean")
).sort_values("Total_Revenue", ascending=False)

top_hospital = hospital_analysis.index[0]
top = hospital_analysis.iloc[0]

# Q2. Which department received the largest number of patients?

department_patients = (
    df.groupby("Department")["Patient_ID"]
    .count()
    .sort_values(ascending=False)
)

highest_department = department_patients.index[0]
highest_department_count = department_patients.iloc[0]
```

Q3. Which disease required the longest recovery period?

Find the actual patient with maximum recovery days.

```
longest_recovery = df.loc[df["Recovery_Days"].idxmax()]
```

Q4. Which admission type resulted in the longest hospital stay?

Find the actual patient with maximum hospital stay.

```
longest_stay = df.loc[df["Hospital_Stay_Days"].idxmax()]
```

Q5. Which hospital achieved the highest patient satisfaction?

```
satisfaction = (  
    df.groupby("Hospital")["Patient_Satisfaction"]  
        .mean()  
        .sort_values(ascending=False)  
)
```

```
highest_satisfaction_hospital = satisfaction.index[0]
```

```
highest_satisfaction = satisfaction.iloc[0]
```

Q6. Which insurance provider covered the largest number of patients?

```
insurance_count = (  
    df.groupby("Insurance")["Patient_ID"]  
        .count()  
        .sort_values(ascending=False)  
)
```

```
highest_insurance = insurance_count.index[0]
```

```
highest_insurance_count = insurance_count.iloc[0]
```

Q7. Which doctor treated the highest number of patients?

```
doctor_count = (  
    df.groupby("Doctor")["Patient_ID"]  
        .count()  
        .sort_values(ascending=False)  
)
```

```

highest_doctor = doctor_count.index[0]
highest_doctor_count = doctor_count.iloc[0]

# Q8. Which hospital spent the most on medicines and laboratory tests?

hospital_expense = df.groupby("Hospital").agg(
    Medicine_Expense=("Medicine_Cost", "sum"),
    Laboratory_Expense=("Lab_Test_Cost", "sum")
)

hospital_expense["Total_Expense"] = (
    hospital_expense["Medicine_Expense"] +
    hospital_expense["Laboratory_Expense"]
)

hospital_expense = hospital_expense.sort_values(
    "Total_Expense", ascending=False
)

highest_expense_hospital = hospital_expense.index[0]
highest_expense = hospital_expense.iloc[0]["Total_Expense"]

# Q9. What percentage of patients were admitted under each department?

department_percentage = (
    df["Department"]
    .value_counts(normalize=True)
    .mul(100)
    .round(2)
)

# Convert percentage values into a clean text line
department_percentage_text = " | ".join(
    [f"{dept}: {pct:.2f}%" for dept, pct in department_percentage.items()]
)

# Q10. Recommendations to hospital management

print("=" * 75)
print("PRACTICAL 3 - RESULTS")

```

```

print("=" * 75)

print("\nQ1. Highest Revenue Hospital")
print(f"Hospital: {top_hospital}")
print(f"Revenue: Rs. {top['Total_Revenue']:.2f}")
print(
    f"Factors: {int(top['Total_Patients'])} patients, "
    f"average treatment cost Rs. {top['Average_Treatment_Cost']:.2f}, "
    f"average surgery cost Rs. {top['Average_Surgery_Cost']:.2f}, "
    f"average medicine cost Rs. {top['Average_Medicine_Cost']:.2f}, "
    f"average lab cost Rs. {top['Average_Lab_Cost']:.2f}, "
    f"average stay {top['Average_Hospital_Stay']:.2f} days."
)

print("\nQ2. Department with Largest Number of Patients")
print(f"Department: {highest_department}")
print(f"Patients: {highest_department_count}")

print("\nQ3. Disease with Longest Recovery Period")
print(f"Patient ID: {longest_recovery['Patient_ID']}")
print(f"Disease: {longest_recovery['Disease']}")
print(f"Recovery Period: {longest_recovery['Recovery_Days']} days")

print("\nQ4. Admission Type with Longest Hospital Stay")
print(f"Patient ID: {longest_stay['Patient_ID']}")
print(f"Admission Type: {longest_stay['Admission_Type']}")
print(f"Hospital Stay: {longest_stay['Hospital_Stay_Days']} days")

print("\nQ5. Hospital with Highest Patient Satisfaction")
print(f"Hospital: {highest_satisfaction_hospital}")
print(f"Average Satisfaction: {highest_satisfaction:.2f}/10")

print("\nQ6. Insurance Provider Covering Most Patients")
print(f"Insurance: {highest_insurance}")
print(f"Patients Covered: {highest_insurance_count}")

print("\nQ7. Doctor Treating Most Patients")
print(f"Doctor: {highest_doctor}")
print(f"Patients Treated: {highest_doctor_count}")

```

```

print("\nQ8. Hospital with Highest Medicine + Laboratory Expense")
print(f"Hospital: {highest_expense_hospital}")
print(f"Total Expense: Rs. {highest_expense:,.2f}")

print("\nQ9. Department-wise Patient Percentage")
print(department_percentage_text)

print("\nQ10. Recommendations")
print(f"1. Allocate more resources to {highest_department}, which has the highest patient load.")
print(f"2. Study the practices of {top_hospital}, the highest revenue-generating hospital.")
print(f"3. Monitor the workload of {highest_doctor}, who treated the most patients.")
print(f"4. Study the practices of {highest_satisfaction_hospital} to improve patient satisfaction.")
print(f"5. Monitor medicine and laboratory expenses, especially at {highest_expense_hospital}.")
print(f"6. Maintain efficient coordination with {highest_insurance} insurance provider.")
print("7. Analyze long recovery periods and hospital stays to improve treatment planning.")

```

Description:

The practical performs hospital patient analytics using the Pandas library. The Hospital Patient Analytics dataset contains 4000 patient admission records and 20 features covering hospitals, departments, doctors, diseases, admission types, insurance providers, treatment costs, hospital stay, recovery period, patient satisfaction and outcomes.

The dataset was grouped using categorical attributes such as Hospital, Department, Disease, Admission Type, Insurance and Doctor. Aggregate functions including sum, mean and count were used to analyze hospital revenue, patient distribution, doctor workload, treatment costs, patient satisfaction and insurance coverage.

The practical also calculates the percentage distribution of patients across departments and performs multi-level GroupBy analysis using Hospital and Department.

For the questions related to the longest recovery period and longest hospital stay, the actual patient record having the maximum number of days was identified. Additional GroupBy analysis was also performed to support categorical analysis.

Output:

Q1. Highest Revenue Hospital

Hospital: AIIMS Delhi

Revenue: Rs. 139,599,517.00

Factors: 717 patients, average treatment cost Rs. 194,699.47, average surgery cost Rs. 35,032.81, average medicine cost Rs. 51,619.19, average lab cost Rs. 16,593.49, average stay 8.31 days.

Q2. Department with Largest Number of Patients

Department: General Medicine

Patients: 989

Q3. Disease with Longest Recovery Period

Patient ID: P03210

Disease: Lung Cancer

Recovery Period: 45 days

Q4. Admission Type with Longest Hospital Stay

Patient ID: P00538

Admission Type: Emergency

Hospital Stay: 21 days

Q5. Hospital with Highest Patient Satisfaction

Hospital: Fortis Hospital

Average Satisfaction: 9.41/10

Q6. Insurance Provider Covering Most Patients

Insurance: Private

Patients Covered: 1849

Q7. Doctor Treating Most Patients

Doctor: Dr. Joshi

Patients Treated: 255

Q8. Hospital with Highest Medicine + Laboratory Expense

Hospital: AIIMS Delhi

Total Expense: Rs. 48,908,491.00

Q9. Department-wise Patient Percentage

General Medicine: 24.72% | Orthopedics: 11.22% | Cardiology: 10.70% | Pediatrics: 9.57% |
Neurology: 8.40% | Gastroenterology: 7.68% | Pulmonology: 7.30% | Nephrology: 6.95% |
Oncology: 6.82% | Dermatology: 6.62%

Q10. Recommendations

1. Allocate more resources to General Medicine, which has the highest patient load.
2. Study the practices of AIIMS Delhi, the highest revenue-generating hospital.
3. Monitor the workload of Dr. Joshi, who treated the most patients.
4. Study the practices of Fortis Hospital to improve patient satisfaction.
5. Monitor medicine and laboratory expenses, especially at AIIMS Delhi.
6. Maintain efficient coordination with Private insurance provider.
7. Analyze long recovery periods and hospital stays to improve treatment planning.

Observations:

- The Hospital Patient Analytics dataset was successfully loaded and analyzed using the Pandas library.
- GroupBy operations were successfully performed on categorical attributes such as Hospital, Department, Disease, Admission Type, Insurance, and Doctor.
- AIIMS Delhi generated the highest revenue. Its high revenue can be attributed to its large number of patients and comparatively high treatment costs.
- General Medicine received the largest number of patients, indicating that it had the highest patient demand among the departments.
- The patient with the longest recovery period was identified, and the corresponding disease and recovery duration were obtained from the dataset.
- The patient with the longest hospital stay was identified along with the corresponding admission type.
- Fortis Hospital achieved the highest average patient satisfaction, indicating comparatively better patient experience.
- Private insurance covered the largest number of patients, making it the most common insurance provider type in the dataset.
- Dr. Joshi treated the highest number of patients, indicating the highest workload among the doctors.
- AIIMS Delhi had the highest combined expenditure on medicines and laboratory tests.
- The percentage distribution of patients across all departments was calculated successfully using percentage-based frequency analysis.
- Additional GroupBy analysis was performed to identify the department with the highest average treatment cost, the department with the highest number of senior citizens, the insurance type contributing the highest treatment revenue, and the department having the highest number of patients under treatment.
- Multi-level GroupBy analysis using Hospital and Department was performed to obtain patient count, total revenue, average treatment cost, and average patient satisfaction.
- The analysis helps hospital management in resource allocation, revenue management, doctor workload management, cost control, insurance management, and improvement of patient satisfaction.
- Overall, the GroupBy and aggregate analysis provided useful insights into hospital performance, patient distribution, treatment costs, doctor workload, recovery trends, and patient care.